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Chance Models in Genetics

I shall never believe that God plays dice with the world.
—ALBERT EINSTEIN (1879–1955)

1. HOW MENDEL DISCOVERED GENES

This chapter is hard, and it can be skipped without losing the thread of the argument in the book. It is included for two reasons:

- Mendel's theory of genetics is great science.
- The theory shows the power of simple chance models in action.

In 1865, Gregor Mendel published an article which provided a scientific explanation for heredity, and eventually caused a revolution in biology.[1] By a curious twist of fortune, this paper was ignored for about thirty years, until the theory was simultaneously rediscovered by three men, Correns in Germany, de Vries in Holland, and Tschermak in Austria. De Vries and Tschermak are now thought to have seen Mendel's paper before they published, but Correns apparently found the idea by himself.

Mendel's experiments were all carried out on garden peas; here is a brief account of one of these experiments. Pea seeds are either yellow or green. (As the phrase suggests, seed color is a property of the seed itself,[2] and not of the parental plant: indeed, one parent often has seeds of both colors.) Mendel bred a pure yellow strain, that is, a strain in which every plant in every generation had

Gregor Mendel (Austria, 1822–1884).
From the collection of the Moravian Museum, Brno.

only yellow seeds. Separately, he bred a pure green strain. He then crossed plants of the pure yellow strain with plants of the pure green strain. For instance, he used pollen from the yellows to fertilize ovules on plants of the green strain. (The alternative method, using pollen from the greens to fertilize plants of the yellow strain, gave exactly the same results.) The seeds resulting from a yellow-green cross, and the plants into which they grow, are called *first-generation hybrids.* First-generation hybrid seeds are all yellow, indistinguishable from seeds of the pure yellow strain. The green seems to have disappeared completely.

These first-generation hybrid seeds grew into first-generation hybrid plants which Mendel crossed with themselves, producing *second-generation hybrid* seeds. Some of these second-generation seeds were yellow, but some were green. So the green disappeared for one generation, but reappeared in the second. Even more surprising, the green reappeared in a definite, simple proportion. Of the second-generation hybrid seeds, about 75% were yellow and 25% were green.

What is behind this regularity? To explain it, Mendel postulated the existence of the entities now called *genes.*[3] According to Mendel's theory, there were two different variants of a gene which paired up to control seed color. They will be denoted here by *y* (for yellow) and *g* (for green). It is the gene-pair in the seed—not the parent—which determines what color the seed will be, and all the cells making up a seed contain the same gene-pair.

There are four different gene-pairs: y/y, y/g, g/y, and g/g. Gene-pairs control seed color by the rule

- y/y, y/g, and g/y make yellow,
- g/g makes green.

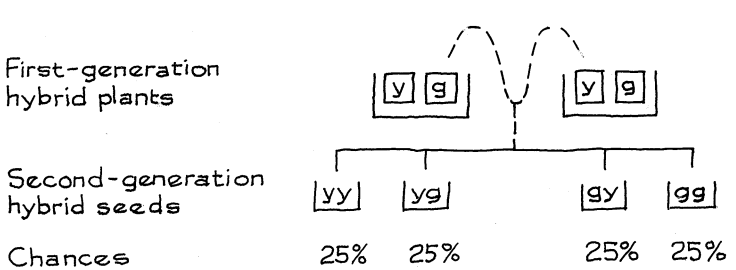
As geneticists say, y is *dominant* and g is *recessive*. This completes the first part of the model.

Now the seed grows up and becomes a plant. All the cells in this plant will also carry the seed's color gene-pair—with one exception. Sex cells, either sperm or eggs, contain only one gene of the pair.[4] For instance, a plant whose ordinary cells contain the gene-pair y/y will produce sperm cells containing the gene y. Similarly, the plant will produce egg cells containing the gene y. On the other hand, a plant whose ordinary cells contain the gene-pair y/g will produce some sperm cells containing the gene y, and some sperm cells containing the gene g. In fact, half its sperm cells will contain y, and the other half will contain g; half its eggs will contain y, the other half will contain g.

This model accounts for the experimental results. Plants of the pure yellow strain have the color gene-pair y/y, so the sperm and eggs all just contain the gene y. Similarly, plants of the pure green strain have the gene-pair g/g, so their pollen and ovules just contain the gene g. Crossing a pure yellow with a pure green amounts for instance to fertilizing a g-egg by a y-sperm, producing a fertilized cell having the gene-pair y/g. This cell reproduces itself and eventually becomes a seed, in which all the cells have the gene-pair y/g and are yellow in color. The model has explained why all first-generation hybrid seeds are yellow, and none are green.

What about the second generation? A first-generation hybrid seed grows into a first-generation hybrid plant, with the gene-pair y/g. This plant produces sperm cells, of which half will contain the gene y and the other half will contain the gene g. The plant also produces eggs, of which half will contain y and the other half will contain g. When two first-generation hybrids are crossed, a resulting second-generation hybrid seed gets one gene at random from each parent—because the seed is formed by the random combination of a sperm cell and an egg. From the point of view of the seed, it's as if one ticket was chosen at random from each of two boxes. In each box, half the tickets are marked y and the other half are marked g. The tickets are the genes, and there is one box for each parent (figure 1).

Figure 1. Mendel's chance model for the genetic determination of seed-color: one gene is chosen at random from each parent. The chance of each combination is shown. (The sperm gene is listed first; in terms of seed color, the combinations y/g and g/y are not distinguishable after fertilization.[5])



As shown in Figure 1, the seed has a 25% chance to get a gene-pair with two g's and be green. The seed has a 75% chance to get a gene-pair with one or two y's and be yellow. The number of seeds is small by comparison with the number of pollen grains, so the selections for the various seeds are essentially independent. The conclusion: the color of second-generation hybrid seeds will be determined as if by a sequence of draws with replacement from the box

yellow	yellow	yellow	green

And that is how the model accounts for the reappearance of green in the second generation, for about 25% of the seeds.

Mendel made a bold leap from his experimental evidence to his theoretical conclusions. His reconstruction of the chain of heredity was based entirely on statistical evidence of the kind discussed here. And he was right. Modern research in genetics and molecular biology is uncovering the chemical basis of heredity, and has provided ample direct proof for the existence of Mendel's hypothetical entities. As we know today, genes are segments of DNA on chromosomes—the dark patches in Figure 2 on the next page.

Essentially the same mechanism of heredity operates in all forms of life, from dolphins to fruit flies. So the genetic model proposed by Mendel unlocks one of the great mysteries of life. How is it that a pea-seed always produces a pea, and never a tomato or a whale? Furthermore, the answer turns out to involve chance in a crucial way, despite Einstein's quote at the opening of the chapter.

Exercise Set A

1. In some experiments, a first-generation hybrid pea is "back-crossed" with one parent. If a y/g plant is crossed with a g/g, about what percentage of the seeds will be yellow? Of 1,600 such seeds, what is the chance that over 850 will be yellow?

2. Flower color in snapdragons is controlled by one gene-pair. There are two variants of the gene, r (for red) and w (for white). The rules are:

   r/r makes red flowers,
   r/w and w/r make pink flowers,
   w/w makes white flowers.

   So neither r nor w is dominant. Their effects are *additive*, like mixing red paint with white paint.

   (a) Work out the expected percentages of red-, pink-, and white-flowered plants resulting from the following crosses: white × red, white × pink, pink × pink.
   (b) With 400 plants from pink × pink crosses, what is the chance that between 190 and 210 will be pink-flowered?

3. Snapdragon leaves come in three widths: wide, medium, and narrow. In breeding trials, the following results are obtained:

   wide × wide → 100% wide
   wide × medium → 50% wide, 50% medium
   wide × narrow → 100% medium
   medium × medium → 25% narrow, 50% medium, 25% wide.

[Exercise continues on p. 463.]

Figure 2. Photomicrograph. These cells are from the root tip of a pea plant, and are magnified about 2,000 times. The cell shown in the center is about to divide. At this stage, each individual chromosome consists of two identical pieces, lying side by side. There are fourteen chromosomes arranged in seven homologous pairs, indicated by the Roman numerals from I to VII. The gene-pair controlling seed-color is located on chromosome pair I, one of the genes being on each chromosome.[6]

Source: New York State Agriculture Experiment Station, Geneva, N.Y.

(a) Can you work out a genetic model to explain these results?

(b) What results would you expect from each of the following crosses: narrow × narrow, narrow × medium?

4. Eye color in humans is determined by one gene-pair, with brown dominant and blue recessive. In a certain family, the husband had a blue-eyed father; he himself has brown eyes. The wife has blue eyes. They plan on having three children. What is the chance that all three will have brown eyes? (It is better to work this out exactly rather than using the normal approximation.)

The answers to these exercises are on p. A90.

2. DID MENDEL'S FACTS FIT HIS MODEL?

Mendel's discovery ranks as one of the greatest in science. Today, his theory is amply proved and extremely powerful. But how good was his own experimental proof? Did Mendel's data prove his theory? Only too well, answered R. A. Fisher:

> ...the general level of agreement between Mendel's expectations and his reported results shows that it is closer than would be expected in the best of several thousand repetitions. The data have evidently been sophisticated systematically, and after examining various possibilities, I have no doubt that Mendel was deceived by a gardening assistant, who knew only too well what his principal expected from each trial made.[7]

Leave the gardener aside for now. Fisher is saying that Mendel's data were fudged. The reason: Mendel's observed frequencies were uncomfortably close to his expected frequencies, much closer than ordinary chance variability would permit.

In one experiment, for instance, Mendel obtained 8,023 second-generation hybrid seeds. He expected $1/4 \times 8{,}023 \approx 2{,}006$ of them to be green, and observed 2,001, for a discrepancy of 5. According to his own chance model, the data on seed color are like the results of drawing 8,023 times with replacement from the box

yellow	yellow	yellow	green

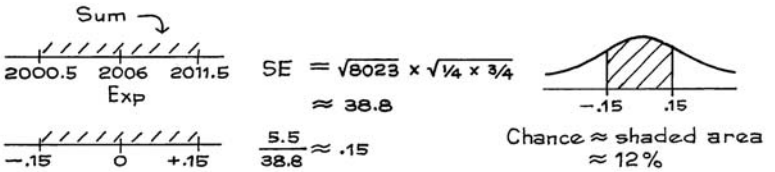
In this model, what is the chance of observing a discrepancy of 5 or less between the number of greens and the expected number? In other words, what is the probability that the number of greens will be

$$\text{between } 1/4 \times 8{,}023 - 5 \approx 2{,}001 \text{ and } 1/4 \times 8{,}023 + 5 \approx 2{,}011?$$

That is like drawing 8,023 times with replacement from the box

0	0	0	1

and asking for the chance that the sum will be between 2,001 and 2,011 inclusive. This chance can be estimated using the normal approximation, keeping track of the edges of the rectangles, as on p. 317.



About 88% of the time, chance variation would cause a discrepancy between Mendel's expectations and his observations greater than the one he reported.

By itself, this evidence is not very strong. The trouble is, every one of Mendel's experiments (with an exception to be discussed below) shows this kind of unusually close agreement between expectations and observations. Using the χ^2-test to pool the results (chapter 28), Fisher showed that the chance of agreement as close as that reported by Mendel is about four in a hundred thousand. To put this another way, suppose millions of scientists were busily repeating Mendel's experiments. For each scientist, imagine measuring the discrepancy between his observed frequencies and the expected frequencies by the χ^2-statistic. Then by the laws of chance, about 99,996 out of every 100,000 of these imaginary scientists would report a discrepancy between observations and expectations greater than the one reported by Mendel. That leaves two possibilities:

- either Mendel's data were massaged
- or he was pretty lucky.

The first is easier to believe.

One aspect of Fisher's argument deserves more attention. However, the discussion is technical, and readers can skip to the beginning of the next section. Mendel worked with six characteristics other than seed color. One of them, for instance, was the shape of the pod, which was either inflated (the dominant form) or constricted (the recessive form). The hereditary mechanism is very similar to that for seed color. Pod shape is controlled by one gene-pair. There are two variants of the shape-gene, denoted by i (inflated) and c (constricted). The gene i is dominant, so i/i or i/c or c/i make inflated pods, and c/c makes constricted pods. (The gene-pair controlling seed color acts independently of the pair controlling pod shape.)

There is one difference between seed color and pod shape. Pod shape is a characteristic of the parent plant, and is utterly unaffected by the fertilizing pollen. Thus, if a plant of a pure strain showing the recessive constricted form of seed pods is fertilized with pollen from a plant of pure strain showing the dominant inflated form, all the resulting seed pods will have the recessive constricted form. But when the seeds of this cross grow up into mature first-generation hybrid plants and make their own seed pods, they will all exhibit the dominant inflated form.

If first-generation hybrids are crossed with each other, of the second-generation hybrid plants about 3/4 will exhibit the dominant form and 1/4 the recessive form. As Figure 1 shows, of the second-generation hybrid plants

with the dominant inflated form, about

$$\frac{25\%}{25\% + 25\% + 25\%} = \frac{1}{3}$$

should be i/i's and the other 2/3 should be i/c or c/i. Mendel checked this out on 600 plants, finding 201 i/i's, a result too close to the expected 200 for comfort.[8] (The chance of such close agreement is only 10%.)

But worse is yet to come. You can't tell the i/i's from the i/c's or c/i's just by looking, the appearances are identical. So how did Mendel classify them? Well, if undisturbed by naturalists, a pea plant will pollinate itself. So Mendel took his second-generation hybrid plants showing the dominant inflated form, and selected 600 at random. He then raised 10 offsprings from each of his selected plants. If the plant bred true and all 10 offsprings showed the dominant inflated form, he classified it as i/i. If the plant produced any offspring showing the recessive constricted form, he classified it as i/c or c/i.

There is one difficulty with this scheme, which Mendel seems to have overlooked. As Figure 1 shows, the chance that the offspring of a self-fertilized i/c will contain at least one dominant gene i, and hence show the dominant inflated form, is 3/4. So the chance that 10 offsprings of an i/c crossed with itself will all show the dominant form is $(3/4)^{10} \approx 6\%$. Similarly for c/i's. The expected frequency of plants classified as i/i is therefore a bit higher than 200, because about 6% of the 400 i/c's and c/i's will be incorrectly classified as i/i. Indeed, the expected frequency of plants classified as i/i—correctly or incorrectly—is

$$200 + 0.06 \times 400 = 224.$$

Mendel's observed frequency (201 classified as i/i) is rather too far from expectation: the chance of such a large discrepancy is only about 5%. As Fisher concludes, "There is no easy way out of the difficulty."

3. THE LAW OF REGRESSION

This section is difficult, and readers can skip to the next section. Part III discussed Galton's work on heredity, and presented his finding that on the average a child is halfway between the parent and the average. In 1918, Fisher proposed a chance model[9] based on Mendel's ideas, which explained Galton's finding on regression as well as the approximate normality of many biometric characteristics like height (chapter 5). The model can be made quite realistic at the expense of introducing complications. This section begins with a stripped-down version which is easier to understand; later, some refinements will be mentioned. The model will focus on heights, although exactly the same argument could be made for other characteristics. The first assumption in the model is

(1) height is controlled by one gene-pair.

The second assumption is

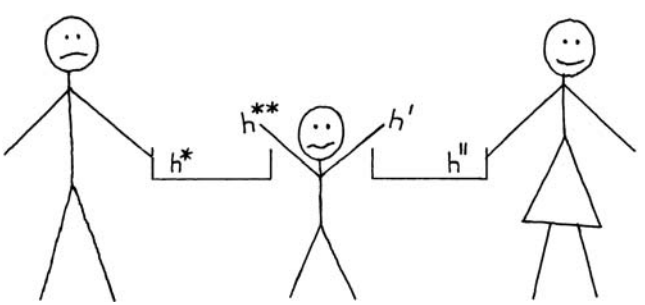
(2) the genes controlling height act in a *purely additive* way.

The symbols h^*, h^{**}, h', h'' will be used to denote four typical variants of the height-gene. (Variants of a gene are called "alleles.") Assumption (2) means, for instance, that h^* always contributes a fixed amount to an individual's height, whether it is combined with another h^*, or with an h', or with any other variant of the height gene. These genes act very differently from the y's and g's controlling seed color in Mendel's peas: g contributes green to the seed color when it is combined with another g, but when it is combined with a y it has no effect. The height genes are more like the snapdragon genes in exercises 2 and 3 on p. 461 above.

With assumption (2), each gene contributes a fixed amount to an individual's height. This contribution (say in inches) will be denoted by the same letter as used to denote the gene, but in capitals. Thus, an individual with the gene-pair h^*/h' will have height equal to the sum $H^* + H'$. In the first instance, the letters refer to the genes; in the second, to the contributions to height.

Fisher assumed with Mendel that a child gets one gene of the pair controlling height at random from each parent (figure 3). To be more precise, the father has a gene-pair controlling height, and so does the mother. Then one gene is drawn at random from the father's pair and one from the mother's pair to make up the child's pair.

Figure 3. The simplified Mendel-Fisher model for the genetic determination of height. Height is controlled by one gene-pair, with purely additive genetic effects. One gene is drawn at random from each parent's gene-pair to make up the child's gene-pair.



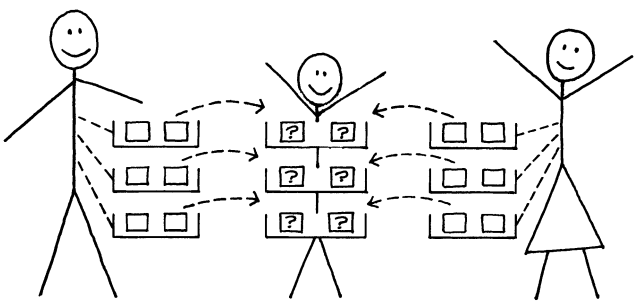
For the sake of argument, suppose the father has the gene-pair h^*/h^{**}, and the mother has the gene-pair h'/h''. The child has chance $1/2$ to get h^* and chance $1/2$ to get h^{**} from the father. Therefore, the father's expected contribution to the child's height is $1/2\,H^* + 1/2\,H^{**} = 1/2\,(H^* + H^{**})$, namely one-half the father's height. Similarly, the mother's expected contribution equals one-half her height. If you take a large number of children of parents whose father's height is fixed at one level, and mother's height is fixed at another level, the average height of these children must be about equal to

(3) $1/2$(father's height + mother's height).

The expression (3) is called the *mid-parent height*. For instance, with many families where the father is 72 inches tall and the mother is 68 inches tall, the mid-parent height is $\frac{1}{2}(72 + 68) = 70$, and on the average the children will be about 70 inches tall at maturity, give or take a small chance error. This is the biological explanation for Galton's law of regression to mediocrity (pp. 169–173).

The assumption (1), that height is controlled by one gene-pair, isn't really needed in the argument; it was made to avoid complicated sums. If three gene-pairs are involved, you only have to assume additivity of the genetic effects and randomness in drawing one gene from each pair for the child (figure 4).

Figure 4. The simplified Mendel-Fisher model for the genetic determination of height, assuming three gene-pairs with purely additive effects. One gene is drawn at random from each gene-pair of each parent to make up the corresponding gene-pair of the child.



So far, the model has not taken into account sex differences in height. One way to get around this is by "adjusting" women's heights, increasing them by around 8% so that women are just as tall as men—at least in the equations of the model. More elegant (and more complicated) methods are available too.

How well does the model fit? For the Pearson–Lee study (p. 119), the regression of son's height on parents' heights was approximately[10]

$$\text{(4)} \qquad \text{estimated son's ht.} = 15'' + 0.8 \times \frac{\text{father's ht.} + 1.08 \times \text{mother's ht.}}{2}$$

The regression coefficient of 0.8 is noticeably lower than the 1.0 predicted by a purely additive genetic model. Some of the discrepancy may be due to environmental effects, and some to nonadditive genetic effects. Furthermore, the sons averaged 1 inch taller than the fathers. This too cannot be explained by a purely additive genetic model.[11]

The regression of son's height on father's height was very nearly

$$\text{(5)} \qquad \text{estimated son's height} = 35'' + 0.5 \times \text{father's height.}$$

Equation (5) can be derived from equation (3) in the additive model, by assuming that there is no correlation between the heights of the parents.[12] Basically, however, this is a case of two mistakes cancelling. The additive model is a bit off,

and the heights of parents are somewhat correlated; but these two facts work in opposite directions, and balance out in equation (5).

Technical note. To derive equation (3) from the model, no assumptions are necessary about the independence of draws from different gene-pairs; all that mattered was each gene having a 50% chance to get drawn. No assumptions are necessary about statistical relationships between the genes in the different parents (such as independence). And no assumptions are necessary about the distribution of the genes in the population (like equilibrium).

4. AN APPRECIATION OF THE MODEL

Genetics represents one of the most satisfying applications of statistical methods. To review the development, Mendel found some striking empirical regularities—like the reappearance of a recessive trait in one-fourth of the second-generation hybrids. He made up a chance model involving what are now called genes to explain his results. He discovered these entities by pure reasoning—he never saw any. Independently, Galton and Pearson found another striking empirical regularity: on the average, a son is halfway between his father and the overall average for sons.

At first sight, the Galton-Pearson results look very different from Mendel's, and it is hard to see how they can both be explained by the same biological mechanism. But Fisher was able to do it. He explained why the average height of children equaled mid-parent height, and even why there are deviations from average. These deviations are caused by chance variation, when genes are chosen at random to pass from the parents to the children.

Chance models are now used in many fields. Usually, the models only assert that certain entities behave as if they were determined by drawing tickets at random from a box, and little effort is spent establishing a physical basis for the claim of randomness. Indeed, the models seldom say explicitly what is like the box, or what is like the tickets.

The genetic model is quite unusual, in that it answers such questions. There are two main sources of randomness in the model:

(i) the random allotment of chromosomes (one from each pair) to sex cells;

(ii) the random pairing of sex cells to produce the fertilized egg.

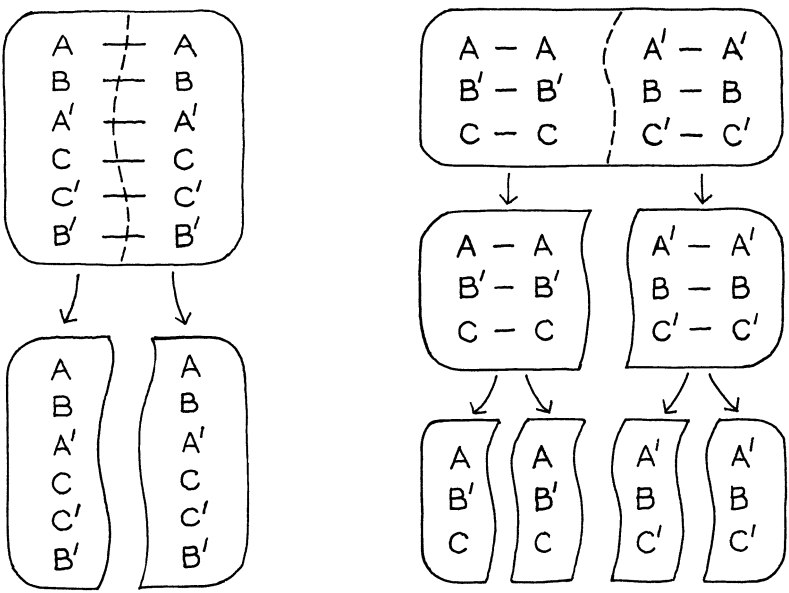
The two sources of randomness will now be discussed in more detail.

Chromosomes naturally come in *homologous* pairs. The one matching C is denoted C'; the chromosomes C and C' are similar, but not identical. A gene-pair has one gene located on each chromosome of a homologous chromosome-pair. A body cell can divide to form other cells. As a preliminary step, each chromosome in the parent cell doubles itself, as shown in figure 2 and (schematically) in figure 5. When chromosome C is in this doubled condition, it will be denoted C-C. The two pieces are chemically identical and loosely joined together.

The production of ordinary body cells is shown in figure 5a. The parent cell splits in two. Each fragment becomes a separate cell with one-half of each doubled chromosome, winding up with exactly the same complement of chromo-

Figure 5. Production of sex-cells and body-cells by splitting. Chromosomes are denoted here by capital letters, like C. Chromosomes come in homologous pairs, as indicated by primes. Thus, C and C' form a homologous pair; they are chemically similar but not identical. As a preliminary to splitting, a cell doubles all its chromosomes. When doubled, C will be denoted by C-C. The two pieces are chemically identical, and loosely attached. Similarly, C' doubles to C'-C'.[13]

(a) Splitting to make body cells (b) Splitting to make sex cells



somes as the parent cell (before doubling). There is nothing random about the resulting chromosomes—it is a matter of copying the whole set. Homologous chromosomes are not treated in any special way.

The production of sex cells is shown in figure 5b. The doubled chromosomes move into position, with one doubled chromosome from each homologous pair on opposite sides of the line along which the cell will split (top of figure 5b). Which side of the line? This seems to be random, like coin-tossing. Sometimes one side, sometimes the other, just as a coin sometimes lands heads, sometimes tails. In the model, the choice of side is assumed to be random.

The cell then splits as shown in the middle of figure 5b. Each fragment contains doubled chromosomes—but only one chromosome of each homologous pair is represented. Finally, each of these fragments splits again, as shown at the bottom of figure 5b, and the results of the second split are the sex cells.[14] The lining-up of the homologous pairs (top of figure 5b) is a critical step. The sex cell contains ordinary, undoubled chromosomes—but only one chromosome out of each homologous pair. Which one? One chosen at random. This is one physical source of randomness in Mendelian genetics.

A fertilized egg results from the union of one male sex cell and one female, out of the many which are produced. Which ones? This seems to be random, like

drawing tickets at random from a box. In the model, the pairing is assumed to be random. This is the second main physical source of randomness in Mendelian genetics.

When thinking about any other chance model, it is good to ask two questions:

- What are the physical entities which are supposed to act like the tickets and the box?
- Do they really act like that?

5. REVIEW EXERCISES

1. Mendel discovered that for peas, the unripe pods are green or yellow. Their color is controlled by one gene-pair, with variants g for green and y for yellow, g being dominant. In a set of breeding trials, plants with known pod color but unknown genetic makeup were crossed. The results are tabulated below.[15] For each line of the table, guess the genetic makeup of the parents:
   (i) g/g (ii) y/g or g/y (iii) y/y

Pod color of parents	*Number of progeny with*	
	green pods	*yellow pods*
green × yellow	82	78
green × green	118	39
yellow × yellow	0	50
green × yellow	74	0
green × green	90	0

2. Mendel found that pea seeds were either smooth or wrinkled. He bred a pure smooth strain and a pure wrinkled strain. Interbreeding these two strains gave first-generation hybrids, which all turned out to be smooth. Mendel crossed the first-generation hybrids with themselves to get second-generation hybrids; of 7,324 second-generation hybrid plants, 5,474 turned out to be smooth, and 1,850 were wrinkled. Make up a genetic model to account for these results. In the model, what is the chance of agreement between the expected frequency of smoothies and the observed frequency as close as that reported by Mendel?

3. Peas flower at three different times: early, intermediate, and late.[16] Breeding trials gave the following results:

   early × early → early
   early × late → intermediate
   late × late → late.

   Suppose you have 2,500 plants resulting from the cross

   intermediate × intermediate.

   What is the chance that 1,300 or more are intermediate-flowering?

4. In humans, there is a special chromosome-pair which determines sex. Males

have the pair X-Y, while females have the pair X-X. A child gets one X-chromosome automatically, from the mother; from the father, it has half a chance to get an X-chromosome and be female, half a chance to get Y and be male. Some genes are carried only on the X-chromosome: these are said to be *sex-linked*. An example is the gene for male-pattern baldness. (Color blindness and hemophilia are other sex-linked characteristics; the model for baldness is simplified.)

(a) If a man has a bald father, is he more likely to go bald?
(b) If a man's maternal grandfather was bald, is he more likely to go bald?

Explain briefly.

5. Sickle-cell anemia is a genetic disease. In the U.S., it is especially prevalent among blacks: one person in four hundred suffers from it. The disease is controlled by one gene-pair, with variants A and a, where a causes the disease but is recessive:

A/A, A/a, a/A—healthy person
a/a—sickle-cell anemia.

(a) Suppose one parent has the gene-pair A/A. Can the child have sickle-cell anemia? How?
(b) Suppose neither parent has sickle-cell anemia. Can the child have it? How?
(c) Suppose both parents have sickle-cell anemia. Can the child avoid having it? How?

6. SUMMARY AND OVERVIEW

1. Whenever reproduction is sexual, the mechanism of heredity is based on gene-pairs. The offspring gets one gene of each pair drawn at random from the corresponding pair in the maternal organism, and one at random from the corresponding pair in the paternal organism. The two genes in a pair are very similar, but not identical.

2. Gene-pairs can control biological characteristics in several ways. One way is *dominance*. In this case, there may be only two varieties (alleles) of the gene, say d and r. The gene-pairs d/d, d/r, r/d all produce the dominant characteristic, while r/r produces the recessive characteristic. (Seed color in peas is an example.) Another way is *additivity*. In this case, each variety of the gene has an effect, and the effect of the gene-pair is the sum of the individual effects of the two genes in the pair. (Flower color in snapdragons is an example.)

3. Fisher showed that Galton's law of regression was a mathematical consequence of Mendel's rules, assuming additive genetic effects.

4. The genetic model explains (at least part of the reason) why children resemble their parents, and also why they differ.

5. This part of the book discussed two chance models: the Gauss model for measurement error and Mendel's model for genetics. These models show how complicated phenomena can be analyzed using the techniques built up in parts II and IV–VI.

6. Chance models are now used in many fields. Usually, the models only assert that some things behave like tickets drawn at random from a box. The genetic model is unusual, because it establishes a physical basis for the claim of randomness.

7. In the next part of the book, we will look at some of the procedures statisticians use for testing models.